ADVANCED LINEAR ALGEBRA
HOMEWORK 6

ROB TOMPKINS

P.4.9. If $A \in \mathbf{M}_n$ is to be represented as $A = XY^\top$ for some $X, Y \in \mathbf{M}_{n \times r}$, explain why r cannot be smaller than $\mathrm{rank}(A)$.

Solution. $\square$

P.4.10. For $X \in \mathbf{M}_{m \times n}$, $\sqsubseteq(X)$ denote the nullity of X.

(a) Show that $\sqsubseteq(X^\top) = \sqsubseteq(X) + m - n$.

(b) Let $A \in \mathbf{M}_{m \times k}$

Solution. $\square$

P.4.13.

Solution. $\square$

P.4.35.

Solution. $\square$

P.4.38.

Solution. $\square$

Date: February 23, 2026.